helix_design design system

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> Master technical specification — **Volume 4 (Clients)**, nano-detail deep-dive. This document **deepens** the *`helix_design` — the design system* section of the pass-1 client overview [04_UI §3, 03-CLIENT §7] into an implementation-ready specification of the `helix_design` Flutter package: the OpenDesign-backed token substrate (§11.4.162), the Material 3 theme layer, the signature connection-state palette, the five signature components (`ConnectButton`, `StatusChip`, `ExitPicker`, `ShieldIndicator`, `AdaptiveScaffold`), the responsive/adaptive one-tree layout, the motion system, accessibility, and the memory/frame budget. SPEC-ONLY: it describes **what to build and why**, to phase → task → subtask granularity; it does **not** build the product.
>
> **Boundary with sibling docs.** This document **consumes** the FFI `TunnelStatus` / `Shields` / `ExitOption` contract owned by `03-CLIENT §3` (which itself mirrors the core enum frozen in `v02-data-plane/orchestrator-and-state.md §4.1`), and the `TunnelPlatform` event contract owned by `03-CLIENT §4–§5`. It **owns** everything visual: tokens, theme, the signature widgets, the state→color mapping, the adaptive shell. It does **not** own the Riverpod providers (`03-CLIENT §8`), the screens, the FFI internals, or any platform shim. The widgets here are pure functions of the status stream (CI2); the design system never opens a tunnel.
>
> **Evidence base.** Citations inline by id: `[04_UI §N]` = `04_VPN_CLD/HelixVPN-helix-ui-Flutter.md`; `[04_ARCH §N]` = `04_VPN_CLD/HelixVPN-Architecture-Refined.md`; `[03-CLIENT §N]` = `final/03-client-core-and-ui.md`; `[02-ORCH §N]` = `final/v02-data-plane/orchestrator-and-state.md`; `[02-TX §N]` = `final/v02-data-plane/transport-trait.md`; `[SYN §N]` = `v09-research/_SYNTHESIS.md`;

> [05_YBO] = operator mandate. Any claim not grounded in the evidence base is tagged UNVERIFIED per constitution §11.4.6 — never fabricated.

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0. Position, ownership, and invariants

A VPN client's UI has one emotional center of gravity: **am I protected right now?** `helix_design` makes that state instantly legible, then gets out of the way [04_UI §3]. Material 3 is the substrate; brand tokens override the defaults so it never reads as a stock Flutter app (a real risk to design against) [04_UI §3, 03-CLIENT §7]. Per §11.4.162 the token-and-theme system is **OpenDesign**, not ad-hoc CSS or one-off styling.

0.1 What this document owns

#	Contract	Owned here
DS1	**Token substrate** — the OpenDesign-sourced design tokens (palette light+dark, typography, spacing, radius, motion, elevation) + their Flutter projection `HelixTokens`	§1
DS2	**Connection-state palette** — the four signature semantic colors (`stateDisconnected/stateConnecting/`	§2

#	Contract	Owned here
	`stateConnected`/`stateDanger`) and the `colorForStatus(TunnelStatus)` mapping	
DS3	**`HelixTheme`** — the `ThemeExtension` carrying tokens into `ThemeData`, light+dark first-class, per-role brand seed override (Access/Connector/Console)	§3
DS4	**Signature components** — `ConnectButton`, `StatusChip`, `ExitPicker`, `ShieldIndicator`, `AdaptiveScaffold` (+ `NetworkTile`, `PolicyEditor` referenced, owned by their screens' docs)	§5
DS5	**Responsive/adaptive rules** — one widget tree, branch on size never `Platform.isX`, the compact/medium/expanded breakpoints	§6
DS6	**Motion + a11y + i18n discipline** — emphasized-easing-on-state-change, semantic announcements, RTL, reduced-motion	§7–§8

0.2 What this document does NOT own

- The FFI surface (`start`/`stop`/`statusStream`/`exits`/`setShields`), the `TunnelStatus`/`Shields`/`ExitOption` *generation* — owned by `03-CLIENT §3`; this doc consumes the generated Dart mirrors as read-only inputs (§4).
- The `TunnelPlatform` shim contract and per-OS implementations — `03-CLIENT §4–§5`.
- The Riverpod providers, controllers, screens, and `runHelixApp` wiring — `03-CLIENT §6/§8`; this doc supplies the *widgets* those screens compose.
- The core orchestrator `TunnelStatus` enum and its emit discipline — frozen in `02-ORCH §4`; the design system never mutates it, only renders it.

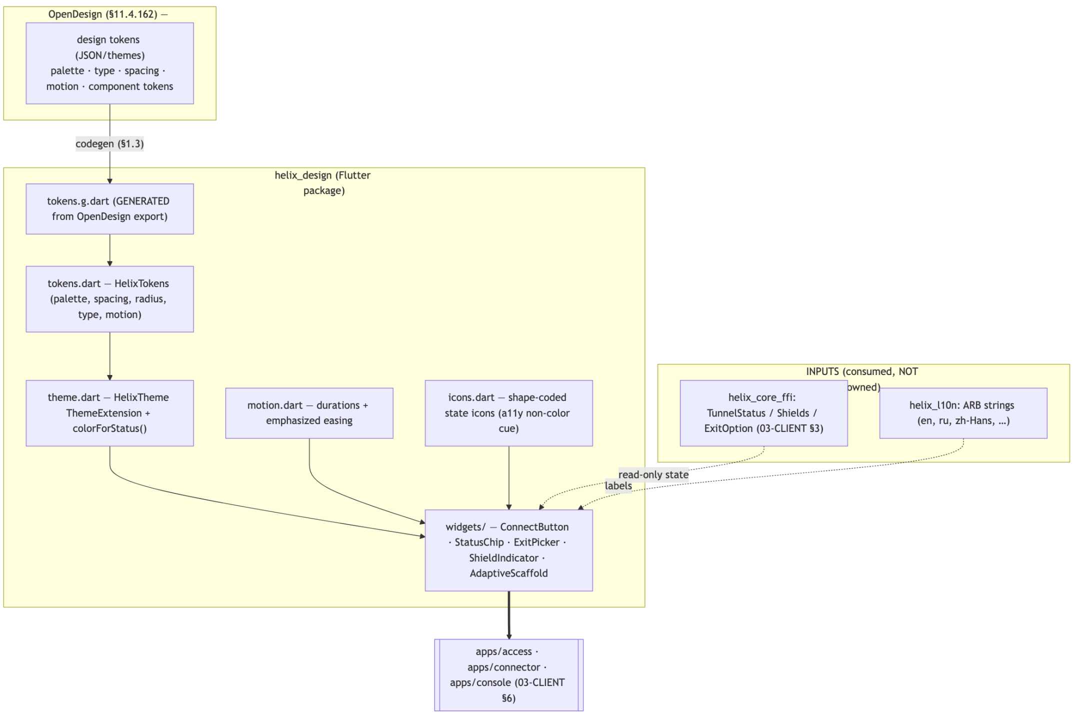
0.3 Invariants this document inherits and tightens

#	Invariant	Source	Design-system tightening
CI2	**The UI is a pure function of the core's status stream.** No polling.	[03-CLIENT §0.1]	Every signature component is a stateless-ish `StatelessWidget` whose visual state is derived **solely** from the

#	Invariant	Source	Design-system tightening
			`TunnelStatus`/ `Shields`/ `ExitOption` it is handed; it holds no independent "connected" belief (§5).
CI5	**State must be announced, not just colored** — never color alone for protected/unprotected.	[04_UI §9, 03-CLIENT §0.1]	Every state-bearing widget pairs its color with a `Semantics(label:)` announcement AND a non-color cue (icon shape / text); a screen-reader user and a colorblind user both get the state (§8).
DS-I1	**Single token source of truth.** Screens reference *roles* (`context.helix.stateConnected`), never hardcoded `Color(0x…)` / magic spacing.	derived [04_UI §3.1]	A future white-label / self-host rebrand is a **token swap, not a refactor**; a hardcoded color in a screen is a release-blocking lint finding (§1.4).
DS-I2	**Light + dark are both first-class**, not an afterthought theme.	[04_UI §9, 03-CLIENT §7.1]	Every component ships and is golden-tested in **both** themes; §11.4.162 mandates light+dark variants per component (§11).
DS-I3	**No element overlaps another; no label overlays a label** (§11.4.162).	§11.4.162	Layout uses constraint-driven composition (`Flex`/ `Wrap`/`OverflowBar`), never absolute `Positioned`

#	Invariant	Source	Design-system tightening
			stacking of text; verified by a visual-regression overlap check (§11).
DS-I4	**Motion is reserved for state change.**	[04_UI §3.3]	The connect transition is the one place to spend animation budget; idle screens are still. Reduced-motion accessibility short-circuits all non-essential animation (§7).

0.4 The package at a glance



`helix_design` is a **decoupled, reusable** package (§11.4.28/§11.4.74) — no project-specific glue, no `helix_core_ffi` dependency in its *code* (it depends only on the *generated mirror types* as a type contract, injected by the consumer). This keeps it independently publishable to its own `vasic-digital` repo [03-CLIENT §2].

1. Token architecture — OpenDesign → Material 3 → HelixTokens

1.1 The three layers

1. **OpenDesign (source of truth, §11.4.162).** OpenDesign supplies the design tokens and themes: the full color palette in **light + dark** variants, the typography scale, the spacing/radius scale, and component-level tokens. Per §11.4.162 the project MUST install OpenDesign as a dependency and use its tokens/themes — NOT ad-hoc CSS/one-off tools. Missing patterns are extended **upstream** per §11.4.74, never duplicated in-project. `UNVERIFIED` (§11.4.6, pending §11.4.99 latest-source verification of `https://github.com/nexu-io/open-design`): OpenDesign's exact Flutter/Dart package name, import path, and token-export file format are not confirmed in the evidence base. This document fixes the **integration contract** (the token taxonomy OpenDesign must supply + the codegen seam) and marks the concrete API binding as an open item (§14 D-DS-1).
2. **Material 3 (Flutter substrate).** OpenDesign tokens are projected onto Flutter's Material 3 `ColorScheme` / `TextTheme` / `ThemeData`. M3 is the widget substrate (ripples, `NavigationBar`, `NavigationRail`, `FilledButton` shapes); the brand tokens override M3 defaults so the app never reads as stock Flutter [04_UI §3, 03-CLIENT §7].
3. **`HelixTokens` (the Dart API screens read).** A flat, `const`, compile-time token surface that screens and widgets reference by *role*. Screens never touch OpenDesign or raw M3 directly (DS-I1).

1.2 `HelixTokens` — the single source of truth

```dart
// helix_design/lib/tokens.dart  — the flat const token surface (DS-I1)
class HelixTokens {
  // ── connection-state palette (the signature semantic colors, §2) ──────────
  static const stateDisconnected = Color(0xFF6B7280); // neutral grey  ← Disconnected
  static const stateConnecting   = Color(0xFFF59E0B); // amber, in-motion ← Connecting/Handshaking/Reconnecting
  static const stateConnected    = Color(0xFF10B981); // green, safe    ← Connected
  static const stateDanger       = Color(0xFFEF4444); // red            ← Danger/Down
```

```
  // ── brand seed: drives the M3 ColorScheme; overridden per role (§3.3) ─────
  static const brandSeed = Color(0xFF3B5BDB);         // HelixVPN brand indigo

  // ── spacing (4-base scale) ──────────────────────────────────────────────
  static const space1 = 4.0, space2 = 8.0, space3 = 12.0, space4 = 16.0,
               space6 = 24.0, space8 = 32.0, space12 = 48.0;

  // ── radius ──────────────────────────────────────────────────────────────
  static const radiusSm = 8.0, radiusMd = 12.0, radiusLg = 20.0, radiusPill = 999.0;

  // ── motion (§7) ─────────────────────────────────────────────────────────
  static const motionFast = Duration(milliseconds: 120);
  static const motionBase = Duration(milliseconds: 220);
  static const motionSlow = Duration(milliseconds: 360);

  // ── elevation: 0..3, used sparingly; flat surfaces, color-as-hierarchy ─────
  static const elevation0 = 0.0, elevation1 = 1.0, elevation2 = 3.0, elevation3 = 6.0;

  // Typography roles map to the M3 scale (display/headline/title/body/label) with
  // the BRAND font (NOT Roboto default); concrete TextStyles live in HelixTheme (§3.2).
}
```

> The literal hex values above are the **pass-1 token values** ratified in [04_UI §3.1, 03-CLIENT §7.1]. Under §11.4.162 these become the HelixVPN *instantiation* of OpenDesign's palette tokens: the codegen (§1.3) emits them into `tokens.g.dart` from the OpenDesign export, and `HelixTokens` re-exports the generated values so the rest of the package stays stable even if OpenDesign's internal naming changes. The brand seed `0xFF3B5BDB` is fixed across all three flavors (the per-role override changes the *derived* scheme, §3.3, not the seed).

1.3 The OpenDesign codegen seam (the no-drift guarantee)

The token pipeline mirrors the project's other codegen drift-gates (`flutter_rust_bridge`, OpenAPI→Dart) [03-CLIENT §11]:

Step	Artifact	Tracked?	Regeneration mechanism (§11.4.77)
1	OpenDesign token export (light+dark)	tracked input	

Step	Artifact	Tracked?	Regeneration mechanism (§11.4.77)
			OpenDesign CLI / package export (`UNVERIFIED` exact cmd → §14 D-DS-1)
2	`tool/gen_tokens.dart`	tracked	hand-authored generator (reads step-1 export → emits step-3)
3	`tokens.g.dart`	tracked (regenerable)	`dart run tool/gen_tokens.dart`
4	drift check	CI/local sweep	`gen_tokens` re-run + `git diff --exit-code tokens.g.dart` — a drift FAILs the build (DS-I1)

A token edit that lands in a screen as a raw `Color(0x…)` or a magic `EdgeInsets` constant (bypassing `HelixTokens`) is caught by a custom `analyzer` lint (`avoid_hardcoded_design_values`, §1.4) — both are DS-I1 violations.

1.4 Lint enforcement of DS-I1 / DS-I3

```
custom_lint rules (helix_design/lint/):
  - avoid_hardcoded_design_values   : flag literal Color()/
EdgeInsets/BorderRadius
                                      outside tokens.dart /
tokens.g.dart
  - require_semantics_on_state_color : a widget reading a state-
palette token MUST
                                      also attach a
Semantics(label:) (CI5, §8)
  - no_text_overlap_stack            : flag a Stack whose children
include >1 Text
                                      without an OverflowBar/Wrap
guard (DS-I3)
```

These are **pre-build gates** (§11.4.4(b) layer 1); each has a paired §1.1 mutation (plant a hardcoded color → lint FAILs; strip a `Semantics` → lint FAILs) per §11.

---

2. The connection-state palette (the signature)

2.1 The four states

The palette is the design system's emotional spine: four semantic colors, each mapping a class of `TunnelStatus` to one instantly-legible meaning [04_UI §3.1].

Token	Hex	Meaning	`TunnelStatus` variant(s)
`stateDisconnected`	`0xFF6B7280` grey	idle, off, safe-to-act	`Disconnected`
`stateConnecting`	`0xFFF59E0B` amber	in-motion, not-yet-protected	`Connecting`, `Handshaking`, `Reconnecting`
`stateConnected`	`0xFF10B981` green	protected, traffic via tunnel	`Connected{transport,path,rtt_ms}`
`stateDanger`	`0xFFEF4444` red	leak / kill-switch tripped / unexpected drop	`Danger{kind}`, `Down{reason}`

2.2 `colorForStatus` — the one mapping the whole app reads

```
// helix_design/lib/theme.dart — the state→color mapping (DS-I1,
      CI5)
extension HelixStatusPalette on BuildContext {
  /// Total over the doc-03 FFI TunnelStatus (§4.1). Every branch
      is covered; an
  /// unmatched variant is a COMPILE error (exhaustive switch) —
      never a silent grey.
  Color colorForStatus(TunnelStatus s) => switch (s) {
    Disconnected()  => HelixTokens.stateDisconnected,
    Connecting() || Handshaking() || Reconnecting() =>
        HelixTokens.stateConnecting,
    Connected()     => HelixTokens.stateConnected,
    Danger() || Down() => HelixTokens.stateDanger,
  };

  /// The NON-COLOR cue paired with every state (CI5: state
      announced, not just colored).
  HelixStateCue cueForStatus(TunnelStatus s) => switch (s) {
    Disconnected()  => HelixStateCue(icon:
        HelixIcons.shieldOff,   label: 'Off'),
    Connecting()    => HelixStateCue(icon:
        HelixIcons.shieldSpin,  label: 'Connecting'),
```

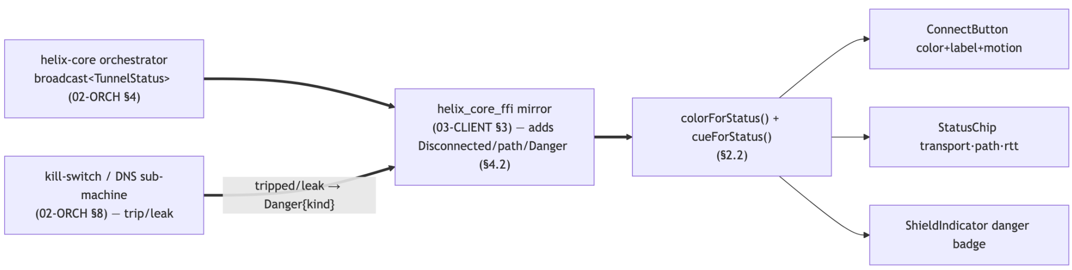
```
    Handshaking()    => HelixStateCue(icon:
        HelixIcons.shieldSpin,  label: 'Securing'),
    Reconnecting()   => HelixStateCue(icon:
        HelixIcons.shieldSpin,  label: 'Reconnecting'),
    Connected()      => HelixStateCue(icon:
        HelixIcons.shieldOn,    label: 'Protected'),
    Down(:final reason)  => HelixStateCue(icon:
        HelixIcons.shieldAlert, label: _downLabel(reason)),
    Danger(:final kind)  => HelixStateCue(icon:
        HelixIcons.shieldAlert, label: _dangerLabel(kind)),
  };
}

class HelixStateCue { final IconData icon; final String label;
  const HelixStateCue({required this.icon, required
        this.label}); }
```

2.3 The danger override (the honesty rule)

`Danger` and `Down{reason≠"stopped"}` **override any user intent** and paint the red palette immediately [03-CLIENT §8.4]. The design system never paints green on intent — only on an actual `Connected` from the core's stream (CI2). The `Danger{kind}` source is the kill-switch/leak sub-machine in `02-ORCH §8` (kill-switch tripped / DNS-leak), synthesized into a `TunnelStatus::Danger` at the FFI boundary (§4.2). `Down.reason` uses the stable-prefix vocabulary (`"stopped"`, `"auth-failed"`, `"ladder-exhausted"`, …) frozen in `02-ORCH §4.4`, so `_downLabel` classifies without parsing prose (§11.4.6).

2.4 Palette ? status flow



2.5 Contrast & accessibility constraint (release-blocking)

Each state color MUST meet **WCAG AA** contrast (⩾ 3:1 for the large `ConnectButton` glyph, ⩾ 4.5:1 for status text) against BOTH the light and dark surface tokens [04_UI §9]. The four-color set is verified at token-build time by a contrast check (§11, `SEC/a11y` test point); a token edit that breaks contrast is a release blocker. The amber/green/grey set was chosen for **colorblind distinguishability** (the shape cue in §2.2 is the fallback for total color deficiency, CI5).

3. HelixTheme — the ThemeExtension

3.1 Why a `ThemeExtension`

Tokens are carried into `ThemeData` via a `ThemeExtension<HelixTheme>` so screens reference roles (`context.helix.stateConnected`) and never hardcode (DS-I1). This keeps the eight platforms visually identical, makes a white-label a token swap, and gives `flutter_test` a single override point for golden tests [04_UI §3.1/§9].

```
// helix_design/lib/theme.dart
@immutable
class HelixTheme extends ThemeExtension<HelixTheme> {
  final Color stateDisconnected, stateConnecting, stateConnected,
        stateDanger;
  final double spaceUnit;                    // 4.0 base; widgets
        multiply
  final HelixMotion motion;                  // durations (§7)
  const HelixTheme({ required this.stateDisconnected, required
        this.stateConnecting,
    required this.stateConnected, required this.stateDanger,
    required this.spaceUnit, required this.motion });

  @override HelixTheme copyWith({ Color? stateDisconnected, /* …
        */ }) => /* … */;
  @override HelixTheme lerp(ThemeExtension<HelixTheme>? other,
        double t) => /* …
    lerp each state color so a theme animation (light↔dark) is
        smooth … */;

  static HelixTheme of(BuildContext c) =>
        Theme.of(c).extension<HelixTheme>()!;
}

extension HelixThemeX on BuildContext {
  HelixTheme get helix => HelixTheme.of(this);   //
        context.helix.stateConnected
}
```

3.2 Theme construction (light + dark, both first-class — DS-I2)

```
// helix_design/lib/theme.dart
ThemeData helixTheme(HelixFlavor flavor, Brightness brightness) {
  final seed = _seedForRole(flavor);                        // §3.3
  final scheme = ColorScheme.fromSeed(seedColor: seed,
        brightness: brightness);
  return ThemeData(
    useMaterial3: true,
```

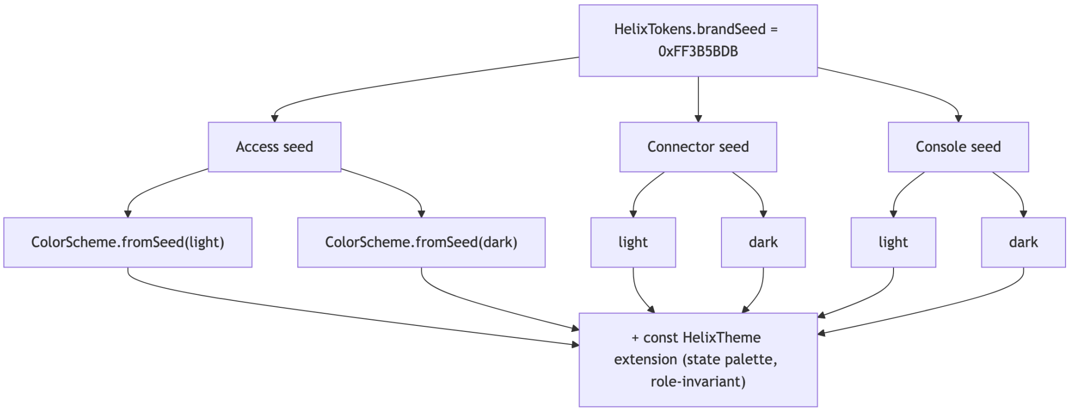
```
    colorScheme: scheme,
    textTheme: _brandTextTheme(scheme),                    // brand font, M3 scale (§1.2)
    extensions: const [ _helixExtension ],                 // const single instance (§9)
    // flat surfaces: low elevation, color-as-hierarchy (04_UI §3.3)
    cardTheme: const CardThemeData(elevation: HelixTokens.elevation0),
    splashFactory: InkSparkle.splashFactory,
  );
}
const _helixExtension = HelixTheme(
  stateDisconnected: HelixTokens.stateDisconnected,
  stateConnecting:   HelixTokens.stateConnecting,
  stateConnected:    HelixTokens.stateConnected,
  stateDanger:       HelixTokens.stateDanger,
  spaceUnit: HelixTokens.space1,
  motion: HelixMotion.standard,
);
```

`light` + `dark` are produced by the *same* call with `Brightness.light/dark`; the state palette is **brightness-invariant** (the four semantic colors are identical in both themes — only the M3 surface/scheme flips), because "am I protected" must read the same regardless of theme. Privacy users skew dark; the system honors the OS brightness with a manual override [04_UI §3.1/§9, 03-CLIENT §7.1].

3.3 Per-role brand override (Access / Connector / Console)

```
Color _seedForRole(HelixFlavor f) => switch (f) {
  HelixFlavor.access    =>
        HelixTokens.brandSeed,              // 0xFF3B5BDB (consumer indigo)
  HelixFlavor.connector =>
        HelixTokens.brandSeed,              // shared seed; role differs by accent
  HelixFlavor.console   =>
        HelixTokens.brandSeed,              // admin; denser tables, same brand
};
```

> `UNVERIFIED` (§11.4.6): the evidence base fixes a **single** brand seed `0xFF3B5BDB` and says it is "overridden per role" [04_UI §3.1, 03-CLIENT §7.1] but does **not** specify distinct per-role seed values. This document therefore ships all three roles on the one ratified seed and marks the *degree* of per-role divergence (a tint shift vs a distinct seed) as an open decision (§14 D-DS-2), rather than fabricate three hex values. The connection-state palette is role-invariant regardless of the outcome (the palette is the safety signal; the brand accent is decoration).



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4. The FFI/platform surface the design system consumes

The design system renders three read-only inputs from the FFI layer. These are **generated mirrors** [03-CLIENT §3.2] — the design system never declares parallel types; it imports the mirror as a type contract.

4.1 `TunnelStatus` (consistent with `02-ORCH §4.1`)

The **core** orchestrator emits the lean five-variant enum frozen in `02-ORCH §4.1` / `02-TX §5.2`:

```
// helix-core/src/status.rs — CANONICAL CORE ENUM (02-ORCH §4.1) —
        do NOT diverge
pub enum TunnelStatus {
    Connecting,
    Handshaking,
    Connected { transport: String, rtt_ms:
        u32 },    // transport = Transport::kind() (02-TX §2)
    Reconnecting,
    Down { reason: String },                    // stable-
        prefix vocabulary (02-ORCH §4.4)
}
```

The **client FFI projection** [03-CLIENT §3.1] mirrors that enum byte-for-byte and **extends** it with three client-only variants/fields the UI needs:

```
// helix-ffi/src/api.rs — the Dart-facing mirror (03-CLIENT §3.1);
        frb generates the Dart twin
#[frb(mirror)]
pub enum TunnelStatus {
    Disconnected,                                              //
        ADDED: clean idle, distinct from Down

        Connecting,                                                //
        —— from here, identical to the core enum ——
```

```
    Handshaking,
    Connected { transport: String, path: String, rtt_ms:
        u32 }, // ADDED field: path = "direct" | "relay"
    Reconnecting,
    Down { reason: String },
    Danger { kind: String },                          //
        ADDED: "leak" | "killswitch_tripped"
}
```

> **Consistency boundary (§11.4.6).** The shared subset {`Connecting`, `Handshaking`, `Connected`, `Reconnecting`, `Down`} is **byte-for-byte the `02-ORCH §4.1` core enum**, and the FFI contract test asserts that match [03-CLIENT §3.2/§12]. The three extensions are synthesized at the FFI boundary, NOT in the core broadcast: `Disconnected` (the FFI's pre-`start`/post-clean-`stop` resting state, where the core has no `OrchState` projection — `02-ORCH §5.1` maps `Idle`/clean-`ShuttingDown` to no public state); `path` on `Connected` (`"direct"` vs `"relay"` — a Phase-2 P2P/relay indicator [03-CLIENT §3.1, §13 T2.3], `UNVERIFIED` that the `02-ORCH` core currently emits it — a cross-doc seam pending the core enum extension); `Danger{kind}` (synthesized from the `02-ORCH §8` kill-switch/DNS sub-machine trip/leak event). The design system renders the 7-variant FFI projection; the *core* contract it must stay consistent with is the 5-variant subset. If the core enum is later extended, the FFI mirror tracks it and the §12 FFI contract test re-asserts byte-equality.

```
// the GENERATED Dart twin the design system imports
        (frb_generated.dart, read-only)
sealed class TunnelStatus { const TunnelStatus(); }
class Disconnected extends TunnelStatus { const Disconnected(); }
class Connecting   extends TunnelStatus { const Connecting(); }
class Handshaking  extends TunnelStatus { const Handshaking(); }
class Connected    extends TunnelStatus {
  final String transport, path; final int rttMs;
  const Connected({required this.transport, required this.path,
        required this.rttMs}); }
class Reconnecting extends TunnelStatus { const Reconnecting(); }
class Down  extends TunnelStatus { final String reason; const
        Down(this.reason); }
class Danger extends TunnelStatus { final String kind; const
        Danger(this.kind); }
```

4.2 How `Danger` / revoked reach the design system across platforms

`Danger{kind}` and the OS-revoked case originate below Dart and must surface to the widgets within the convergence budget [03-CLIENT §4.1 O3]. The design system does not implement this path — it only **consumes** the resulting `TunnelStatus`. The load-bearing native signatures (owned by `03-CLIENT §5`, shown here only to fix where the design-system input comes from):

```
// iOS/macOS NEPacketTunnelProvider (03-CLIENT §5.1) — kill-
        switch/leak → Danger
func emitDanger(_ kind: String) {
        helix_core_emit_status(.danger(kind)) }   // → FFI stream
override func stopTunnel(with reason: NEProviderStopReason, ...)
        { /* revoked → Down/Danger */ }

// Android VpnService (03-CLIENT §5.2)
override fun onRevoke() { coreStop(); emitEvent(REVOKED) }   // →
        PlatformTunnelEvent.revoked → Down

// Aurora Qt/C++ backend (03-CLIENT §5.6) — same shape over the C
        ABI
void HelixTunnelBackend::onLeakDetected() {
        helix_core_emit_status_danger("leak"); }
```

The design system's contract is simply: **whatever `TunnelStatus` the FFI stream emits, render it** (CI2). A `revoked/Danger` arriving while the user *intends* connected paints red, never green (§2.3).

4.3 `Shields` and `ExitOption` (read by `ShieldIndicator` / `ExitPicker`)

```
// generated mirrors (03-CLIENT §3.1) — design system consumes,
        never mutates
class Shields { final bool killSwitch, dnsProtection, daita,
        postQuantum;
  final List<String> splitTunnel; const Shields(/* … */); }
class ExitOption { final String id, kind, label; final String?
        country; final int? rttMs;
  final String? jurisdiction; const ExitOption(/* … */); }
```

`setShields`/`setExit` are pure-logic FFI calls (no OS tunnel, CI1) [03-CLIENT §3.3]; the design-system widgets emit a *callback* (`onToggle`/`onPick`) and never call the FFI directly — the screen's controller does (separation of concerns, §5).

---

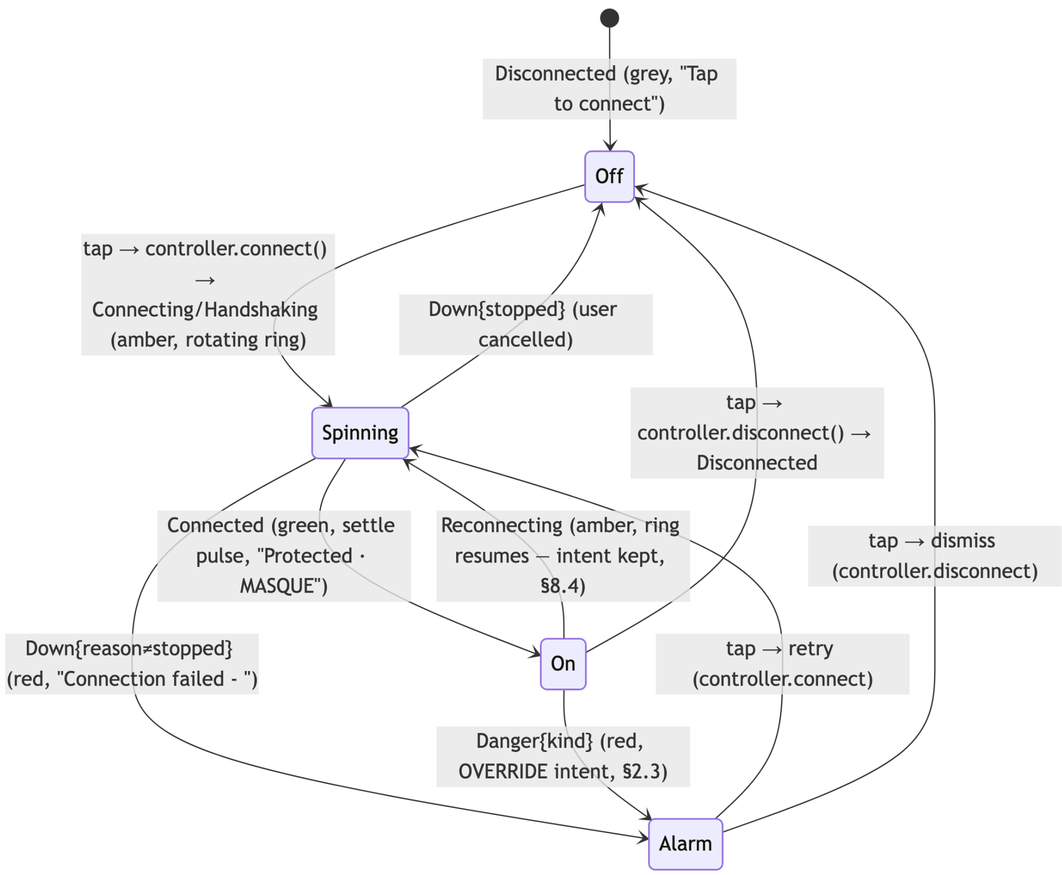
5. Signature components (nano-detail)

Every signature widget obeys five rules: (a) **stateless render from inputs** (no internal "connected" belief, CI2); (b) **color paired with a non-color cue + a `Semantics` announcement** (CI5); (c) **light+dark + every state golden-tested** (DS-I2, §11); (d) **no element/label overlap** (DS-I3); (e) **callbacks out, never FFI calls in** (the widget is pure; the controller acts).

5.1 `ConnectButton` — the giant tap target

```dart
// helix_design/lib/widgets/connect_button.dart
class ConnectButton extends StatelessWidget {
  final TunnelStatus state;            // the ONLY source of
        visual state (CI2)
  final VoidCallback onTap;            // controller decides
        connect vs disconnect (03-CLIENT §8.2)
  final double diameter;              // default 200 (compact) →
        adaptive (§6)
  const ConnectButton({super.key, required this.state, required
        this.onTap,
                       this.diameter = 200});
  @override Widget build(BuildContext c) { /* §5.1.2 */ }
}
```

5.1.1 Visual state machine



5.1.2 Render + motion + a11y

```dart
@override Widget build(BuildContext c) {
  final color = c.colorForStatus(state);
  final cue   = c.cueForStatus(state);
  final spinning = state is Connecting || state is Handshaking ||
        state is Reconnecting;
  return Semantics(
```

```
    button: true,
    label: 'Connection',
    value: cue.label,                                       // CI5:
        announce state to screen reader
    onTapHint: state is Connected ? 'Disconnect' : 'Connect',
    child: GestureDetector(
      onTap: onTap,
      child: RepaintBoundary(                                // §9:
        isolate the animated ring
        child: AnimatedContainer(
          duration:
        c.helix.motion.base,                     // 220 ms, emphasized
        easing (§7)
          curve: HelixCurves.emphasized,
          width: diameter, height: diameter,
          decoration: BoxDecoration(shape: BoxShape.circle,
        color: color.withOpacity(.12),
            border: Border.all(color: color, width: 4)),
          child: Column(mainAxisAlignment:
        MainAxisAlignment.center, children: [
            spinning ? _SpinningRing(color: color) :
        Icon(cue.icon, color: color, size: 48),
            const SizedBox(height: HelixTokens.space2),
            Text(cue.label, style: c.textTheme.titleMedium),  //
        non-color cue (CI5)
          ]),
        ),
      ),
    ),
  );
}
```

- **Edge cases:** a `null`/uninitialized status renders `Disconnected` (the safe default, §10); an unknown `Down.reason` prefix renders the generic "Connection failed" label (never a raw prose dump, §11.4.6); rapid taps are debounced by the controller (the widget just forwards `onTap`).
- **Memory:** the spinning ring is the only animation; it lives behind a `RepaintBoundary` so it never invalidates the rest of the tree (§9). When not spinning, zero `AnimationController` runs (battery on idle).

5.2 `StatusChip` — transport · path · RTT

```
class StatusChip extends StatelessWidget {
  final TunnelStatus state;
  const StatusChip({super.key, required this.state});
  @override Widget build(BuildContext c) {
    final txt = switch (state) {
      Connected(:final transport, :final path, :final rttMs) =>
        '${_label(transport)} · $path · ${rttMs}ms',      //
        "MASQUE · direct · 23ms"
      Connecting() || Handshaking() => 'Connecting…',
      Reconnecting() => 'Reconnecting…',
```

```
      Disconnected() => 'Not connected',
      Down(:final reason) => _downLabel(reason),
      Danger(:final kind) => _dangerLabel(kind),
    };
    return Semantics(label: 'Tunnel status', value: txt,
      child: Chip(label: Text(txt, overflow:
        TextOverflow.ellipsis),  // DS-I3: never overflow-overlap
        avatar: Icon(c.cueForStatus(state).icon, color:
        c.colorForStatus(state), size: 16)));
  }
}
```

- `transport` is the `Transport::kind()` string from `02-TX §2` (`"masque-h3"` → display `"MASQUE"`); `path` is the Phase-2 direct/relay field; `rttMs` is the WG-handshake RTT EWMA, re-emitted only when it drifts > 5 ms [02-ORCH §4.3], so the chip updates without flicker. **Edge case:** an unknown transport string shows the raw label (never a crash), flagged for an `l10n` follow-up (§10).

5.3 `ExitPicker` — exit gateway / network / multi-hop

```
class ExitPicker extends StatelessWidget {
  final List<ExitOption> options;          // from core.exits(),
        RTT-sorted upstream (03-CLIENT §7.2)
  final String? selectedId;
  final ValueChanged<String> onPick;       // callback out
        (controller calls setExit)
  final ExitPickerStatus loadState;        // loading | ready |
        error | empty (§10)
  const ExitPicker({ super.key, required this.options, required
        this.selectedId,
                     required this.onPick, this.loadState =
        ExitPickerStatus.ready });
}
enum ExitPickerStatus { loading, ready, error, empty }
```

- Searchable, RTT-sorted, **jurisdiction labels** for multi-hop chains [04_UI §3.2, 03-CLIENT §7.2]. Each row: `label` · `country` flag · `rttMs` · `jurisdiction` badge. The widget renders rows in a `ListView.builder` (constant memory for a long exit list, §9).
- **Edge cases (§10):** `loading` → shimmer skeletons (not a frozen spinner, the §11.4.107 liveness rule applies to UI loaders too); `empty` → "No exits available"
    - a retry affordance; `error` → honest error text from the FFI failure detail (§11.4.6, never "something went wrong"); a `selectedId` not present in `options` (stale selection after a map reconcile) → clears the selection and surfaces it.
- Multi-hop chain builder (Phase 2 [03-CLIENT §13 T2.1]) composes two `ExitPicker` pickers (entry/exit) with overlap-free jurisdiction labels (DS-I3).

5.4 `ShieldIndicator` — kill-switch / DNS / DAITA / PQ

```
class ShieldIndicator extends StatelessWidget {
  final Shields
        shields;                        // from core (03-CLIENT §3.1)
  final TunnelStatus state;                     // to surface
        Danger{kind} on the relevant badge
  final ValueChanged<Shields> onToggle;     // callback out
        (controller calls setShields)
  const ShieldIndicator({ super.key, required this.shields,
        required this.state,
                          required this.onToggle });
}
```

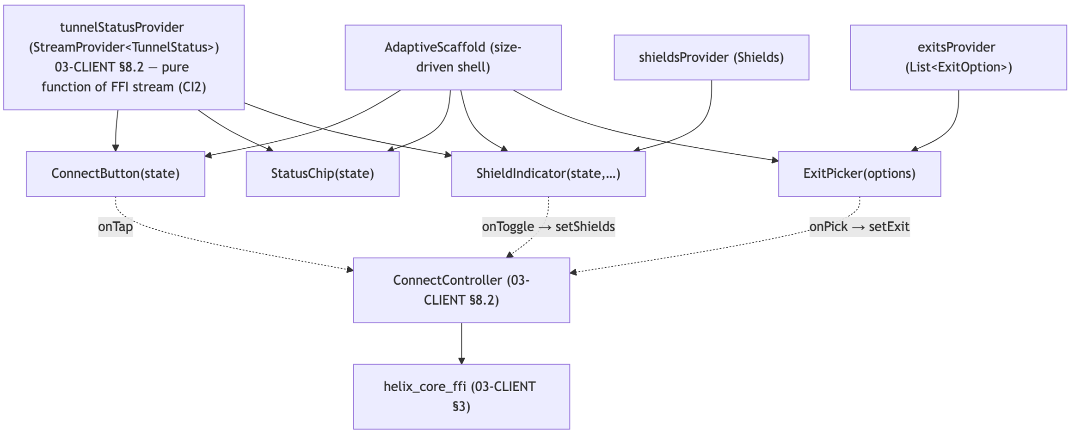
- Renders four badges (`killSwitch`, `dnsProtection`, `daita`, `postQuantum`), each a small affordance with an **honest cost note on tap** ("DAITA adds cover traffic — uses more data") [04_UI §3.2]. A `Danger{kind:"killswitch_tripped"}` paints the kill-switch badge red and announces it (CI5).
- **Honest-cost rule (§11.4.6):** the cost note is factual, never marketing — e.g. post-quantum's note states it is a hybrid PSK that does not slow the data path but adds handshake bytes (the doc-01 PQ stance). No badge ever claims a protection it cannot prove is active; a badge's "on" state is driven by the *core's* `Shields` truth, not the toggle's optimistic position (CI2).

5.5 `AdaptiveScaffold` — the responsive shell

```
class AdaptiveScaffold extends StatelessWidget {
  final List<HelixDestination> destinations;   // (icon, label,
        route)
  final int selectedIndex; final ValueChanged<int> onSelect;
  final Widget body; final Widget? detailPane;  // expanded
        layouts show body|detail
  const AdaptiveScaffold({ super.key, required this.destinations,
        required this.selectedIndex,
                           required this.onSelect, required
        this.body, this.detailPane });
  @override Widget build(BuildContext c) {
    final w = MediaQuery.sizeOf(c).width;         // branch on
        SIZE, never Platform.isX (§6)
    return switch (HelixBreakpoint.of(w)) {
      HelixBreakpoint.compact  => _bottomNavLayout(c),      //
        BottomNavigationBar, single pane
      HelixBreakpoint.medium   => _railLayout(c, extended:
        false),  // NavigationRail, master/detail
      HelixBreakpoint.expanded => _railLayout(c, extended:
        true),   // extended rail, multi-pane
    };
  }
}
```

`AdaptiveScaffold` is the **only** widget that reads `MediaQuery` width; every other widget is layout-agnostic (it fills the constraints it is given), so a resized desktop window and a phone share one code path (§6).

5.6 Component data-flow (all five at once)

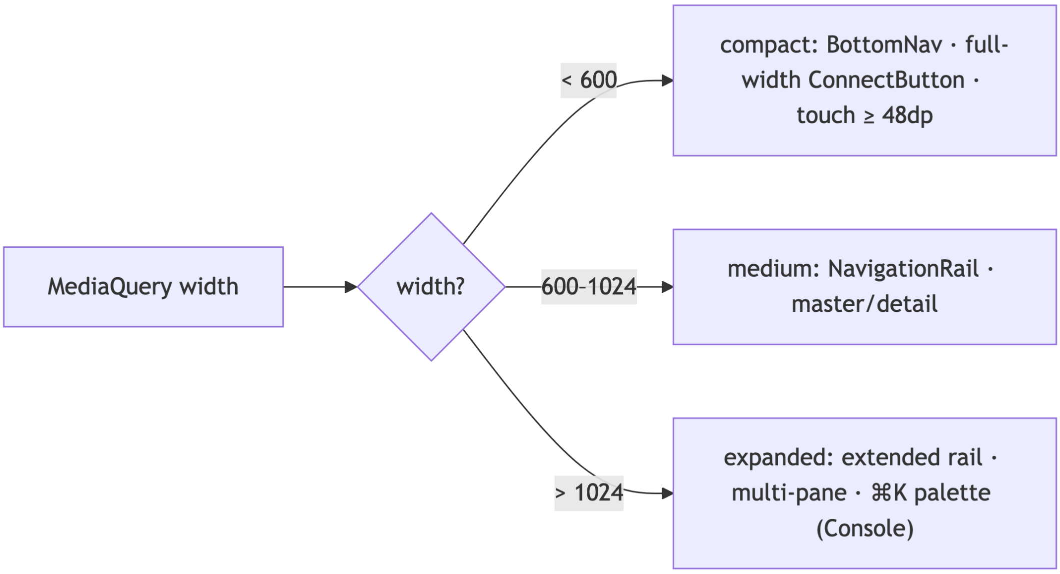


---

6. Responsive / adaptive layout (one tree, every form factor)

One `AdaptiveScaffold` reflows by **width breakpoint** — the convergent approach that makes the same Access app feel native on a phone and a desktop, and the Console feel native on web and desktop [04_UI §7, 03-CLIENT §7.3].

```
// helix_design/lib/widgets/breakpoint.dart
enum HelixBreakpoint {
  compact,   // < 600   : BottomNavigationBar · single pane · full-
        width ConnectButton
  medium,    // 600–1024 : NavigationRail · master/detail
  expanded;  // > 1024 : extended NavigationRail · multi-pane
        (Console: list | detail | live events)
  static HelixBreakpoint of(double w) =>
    w < 600 ? compact : (w <= 1024 ? medium : expanded);
}
```



Rules [04_UI §7, 03-CLIENT §7.3]:

- **Branch on size, never `Platform.isX`** for *layout* — a desktop window resized small behaves like a phone; web "just works" responsively (the brief's "fully responsive web"). `Platform.isX` is reserved for *capability* gating (§11.4.x flavor model, `03-CLIENT §6`), never layout.
- **Touch targets ⩾ 48 dp** on compact; pointer + keyboard shortcuts on medium/expanded (⌘K command palette in Console).
- **Console heavy surfaces** (topology graph, policy editor, dense tables) are **deferred-loaded** so they never bloat Access/Connector binaries [04_UI §7, 03-CLIENT §7.3, §9].
- **No element/label overlap at any width** (DS-I3) — reflow uses `Wrap`/ `Flex`/ `OverflowBar`, never absolute positioning of text; verified by the visual- regression overlap check across all three breakpoints (§11).

---

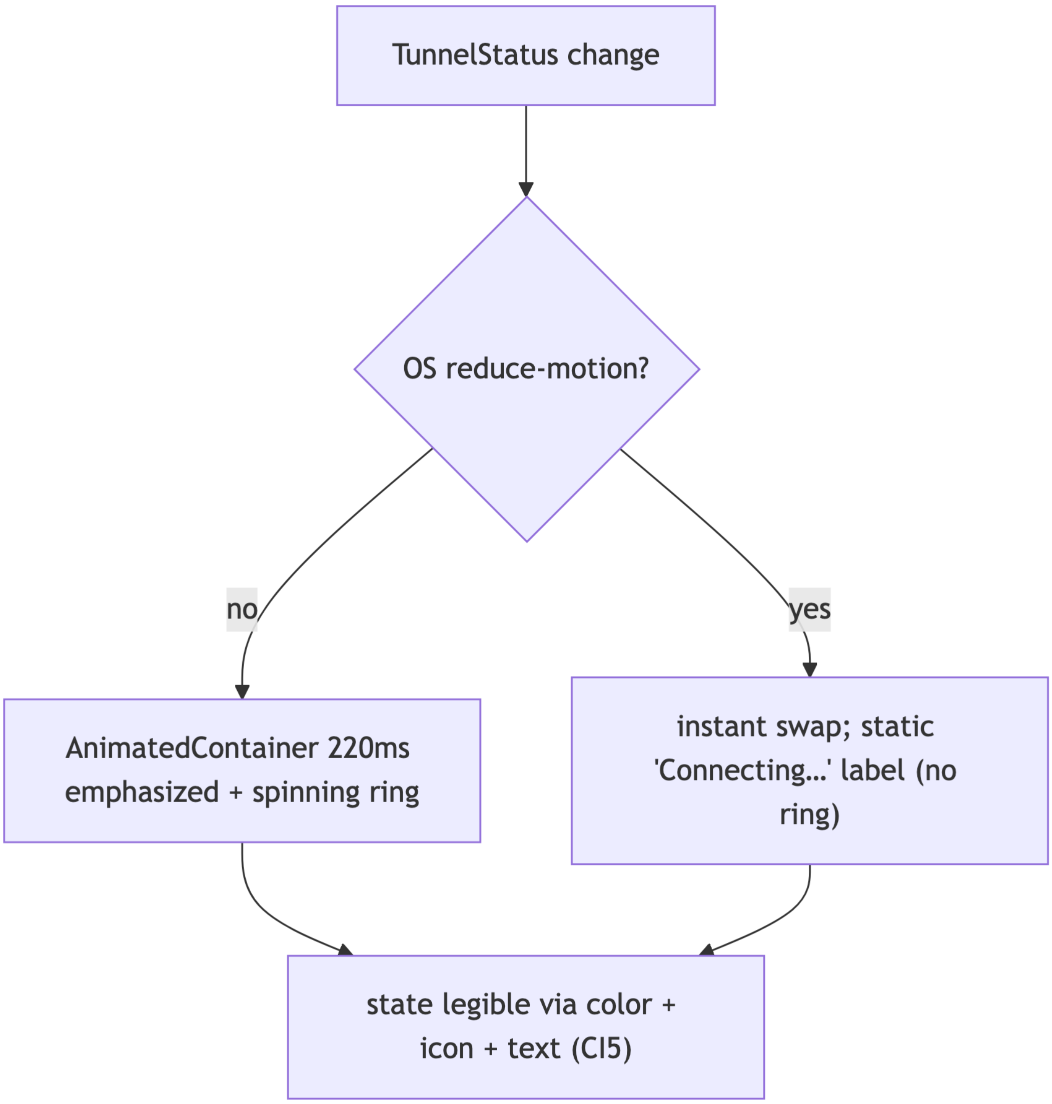
7. Motion system

Motion is **reserved for state change** — the connect transition is the one place to spend the animation budget; idle screens are still (DS-I4) [04_UI §3.3].

```
// helix_design/lib/motion.dart
class HelixMotion {
  final Duration fast, base, slow;
  const HelixMotion({required this.fast, required this.base,
       required this.slow});
  static const standard = HelixMotion(
    fast: HelixTokens.motionFast,    // 120ms – micro-feedback
       (ripples, hover)
    base: HelixTokens.motionBase,    // 220ms – the connect-state
       transition (the spend)
```

```
    slow: HelixTokens.motionSlow);  // 360ms — large surface
        transitions (rare)
}
class HelixCurves {
  static const emphasized = Cubic(0.2, 0.0, 0.0, 1.0);   // M3
        emphasized; state change only
  static const standard   = Curves.easeInOutCubicEmphasized;
}
```

- The `ConnectButton` ring uses `motion.base` + `emphasized` easing on every state transition (§5.1); the settle-pulse on `Connected` is a single `200ms` overshoot, not a loop.
- **Reduced-motion accessibility (release-blocking, CI5):** when `MediaQuery.disableAnimations` (OS "reduce motion") is set, all non-essential animation is short-circuited — the ConnectButton snaps between states (still legible via color+icon+label), the spinning ring becomes a static "Connecting…" label. The state is **never** conveyed by motion alone.



---

8. Accessibility & i18n (CI5)

State must be **announced, not just colored** — never rely on color alone for protected/unprotected [04_UI §9, 03-CLIENT §0.1 CI5]. The design system enforces this mechanically (the `require_semantics_on_state_color` lint, §1.4).

Concern	Rule	Verified by (§11)
Screen reader	every state-bearing widget has `Semantics(label:, value:)`; `ConnectButton` announces "Connection: Protected" + tap hint	a11y test point (`UT`/`E2E`)
Color blindness	every state color paired with a shape-coded icon + text label (§2.2)	golden in a grayscale filter + manual; lint `require_semantics_on_state_color`
Contrast	WCAG AA against light AND dark surfaces (§2.5)	token-build contrast check (release-blocking)
Large text	dynamic type scales without overlap/ clipping (DS-I3); labels `ellipsis`, never truncate-overlap	golden at 2.0× text scale × both themes
Keyboard	full keyboard nav on desktop/web; focus order = visual order; ⌘K palette in Console	`E2E` keyboard-traversal test
RTL / i18n	all strings from `helix_l10n` ARB (en, ru, zh-Hans, …); layout mirrors in RTL; no hardcoded English	golden in `ar`/`he` RTL locale

i18n: all user-facing strings are ARB keys [03-CLIENT §2]; the design system ships **no** literal copy except debug labels. `Down.reason`/`Danger.kind` prefixes (`02-ORCH §4.4`) map to localized strings via a closed lookup (`_downLabel`), never raw prose to the user (§11.4.6).

---

9. Memory / frame budgets

The design system's slice of the doc-03 budgets [03-CLIENT §10, 04_UI §10]:

Budget	Target	How
Frame budget	60/120 fps, no jank	`const` widgets everywhere; `RepaintBoundary` on the ConnectButton ring; motion only on state change (DS-I4)
Idle animation	zero `AnimationController` when not transitioning	the spinning ring stops + disposes its controller on `Connected`/`Disconnected` (§5.1)
Theme allocation	one `const HelixTheme` instance per (flavor × brightness)	`const _helixExtension` (§3.2) — no per-build allocation
Long lists	constant memory for N exits/devices	`ListView.builder`/`Sliver` (lazy), never a materialized `Column` (§5.3)
Console heavy widgets	excluded from Access/ Connector binaries	deferred-loaded (`deferred as`) topology/policy/table widgets (§6)
Token surface	zero runtime cost	`HelixTokens` is all `const`; resolved at compile time (§1.2)

The design system adds **no** Rust/native memory (it is pure Dart/Flutter); the iOS NE ~15 MB working-set ceiling [03-CLIENT §0.1 CI4, §10] is the *core's* budget, untouched by `helix_design`. The design-system contribution to install size is tree-shaken icons/fonts + the brand typeface (one weight set, subset to used glyphs) [03-CLIENT §10].

---

10. Error handling & edge cases

#	Edge case	Behavior (§11.4.6 no-guessing)
E1	FFI stream `Lagged` (broadcast ring fell behind, `02-ORCH §4.2`)	the Riverpod `StreamProvider` skips-to-latest; the widget re-renders the newest `TunnelStatus` — never an error state (§4.6 receiver contract)
E2	`null` / not-yet-initialized status	render `Disconnected` (the safe default, grey, "Not connected") —

#	Edge case	Behavior (§11.4.6 no-guessing)
		never blank, never a spinner-forever
E3	unknown `transport` string in `Connected`	`StatusChip` shows the raw kind label (no crash); flagged for an `l10n` mapping follow-up
E4	unknown `Down.reason` / `Danger.kind` prefix	generic localized "Connection failed" / "Protection alert" label; the raw value is logged (not shown) for forensics (§11.4.6)
E5	`ExitPicker` selectedId not in options (stale after reconcile, `02-ORCH §6.4`)	clear selection + surface "Exit no longer available" (honest, not silent)
E6	`ExitPicker` loading / empty / error	shimmer skeleton / "No exits" + retry / honest FFI error detail — never a frozen spinner (§11.4.107 liveness)
E7	`HelixTheme.of(context)` missing (extension not registered)	assertion failure in debug (fail-loud); release builds fall back to M3 defaults — caught by a startup smoke test
E8	OpenDesign token missing at codegen	`gen_tokens` FAILs with the exact missing key (§1.3) — a build-blocking finding, never a silent default (§11.4.6)
E9	dynamic-type overflow on a small width	text ellipsis + tooltip; layout reflows (DS-I3), never overlaps
E10	rapid connect/disconnect taps	controller debounces (`03-CLIENT §8.2`); the widget forwards `onTap` idempotently (no double-start)
E11	`Danger` arrives while intent = connected	red palette OVERRIDES intent immediately (§2.3); the button shows the alarm state, never green

---

11. Test points — tied to §11.4.169

Every `helix_design` workable item declares its required test types from the **§11.4.169 closed test-type vocabulary**; the only permitted absence of a warranted type is an honest §11.4.3 SKIP-with-reason, never a silent gap.

Four-layer enforcement per §11.4.4(b) applies to every component closure. §11.4.162 adds the **visual-regression** mandate (light+dark × breakpoint × state golden matrix).

Code (§11.4.169)	**Type**	**Concrete test point for `helix_design`**	**Evidence (§11.4.5/.69/.107/.159**
`UT`	unit / widget + golden	each component renders correctly for **every** `TunnelStatus` × {light, dark} × {compact, medium, expanded}; `colorForStatus`/ `cueForStatus` exhaustive switch; `HelixBreakpoint.of` boundaries (599/600/1024/1025)	`flutter_test` + `golden_toolkit` PNG matrix
`IT`	integration	the design system inside a real flavor shell driven by a **fake `HelixCore`** emitting a scripted `TunnelStatus` sequence (same generated mirror types) → widgets reflect each state	scripted-stream render log
`E2E`	end-to-end	real connect journey on a real device: tap `ConnectButton` → `StatusChip` reads `MASQUE · direct · 23ms` → confirmed by OCR/vision validation (§11.4.158/.159)	window-scoped MP4 (`helixvpn---connect---<run-id>.mp4`, §11.4.155) + OCR verdict
`FA`	full-automation (deterministic §11.4.50)	N=3 identical golden runs across the full state×theme×breakpoint matrix produce byte-identical PNGs	3× identical golden artifacts
`CH`	Challenges (challenges submodule)	a Challenge scores the captured connect-journey MP4 + golden matrix, not config	Challenge `result.json`
`HQA`	HelixQA (helix_qa submodule)	autonomous QA session drives connect→shields→exit-pick across a flavor	HelixQA session evidence
`SEC`	security / a11y-contrast	WCAG AA contrast check on the 4 state colors × {light,dark}; no secret in any token; `require_semantics_on_state_color` lint passes	contrast report; lint output
`SC`	stress + chaos (§11.4.85)	rapid `TunnelStatus` flips (1000 transitions, incl. `Connected⍰Danger⍰Reconnecting`) → no dropped frame, no stuck	frame-timeline CSV; recovery trace

Code (§11.4.169)	Type	Concrete test point for `helix_design`	Evidence (§11.4.5/.69/.107/.159
		animation, controller debounce holds	
`CONC`	concurrency	concurrent stream emission + user tap during a transition → no torn render, no double `onTap`	concurrency harness log
`MEM`	memory	no `AnimationController` leak across 1000 connect/disconnect cycles; constant heap for a 500-row `ExitPicker`	DevTools memory timeline
`BENCH`	performance	ConnectButton transition stays within the 16.6 ms (60 fps) / 8.3 ms (120 fps) frame budget; golden matrix render time	frame-time CSV
`RACE`	race / deadlock	provider-override race on theme swap (light⍰dark mid-animation) → no inconsistent `HelixTheme` read	race harness log
`LOAD`	DDoS / load-flood	**SKIP** — §11.4.3: a pure-UI package has no network/request surface to flood; the load surface is the FFI/ control plane (doc 02), tested there	SKIP-with-reason record

> **Anti-bluff coupling (§11.4.107/§11.4.158/§11.4.159).** A green widget golden is **not** proof the user is protected — it proves the design system *renders the state it is handed*. The authoritative end-user evidence is the `E2E` device recording: a window-scoped MP4 of the real connect journey, vision-validated that the `StatusChip` reads the real transport/path/RTT and the `ConnectButton` is green, plus the §11.4.163 media-validation pipeline confirming the recording's on-screen content matches the SPECIFY-phase expected patterns. Fakes (the `HelixCore` double in `IT`/`UT`) use the **same generated mirror types** the FFI produces, so unit tests exercise the real contract; the user-visible claim is only earned by the device recording (§11.4.27/.52).

Each lint/gate (§1.4) carries a paired §1.1 meta-test mutation: plant a hardcoded `Color(0x…)` in a screen → `avoid_hardcoded_design_values` FAILs; strip a `Semantics` from a state widget → `require_semantics_on_state_color` FAILs; seed a golden with an overlapping label → the DS-I3 overlap check FAILs; break a state color's contrast → the `SEC` contrast gate FAILs.

---

12. Phase → task → subtask plan

Aligned to the program roadmap [SYN §4, 03-CLIENT §13]. Each subtask is a workable item (§11.4.93 SQLite SSoT).

Phase 0 — prove the seam (the spike)

- **T0.1 Token + theme skeleton.** `HelixTokens` (the ratified pass-1 values), `HelixTheme` extension, light+dark `helixTheme()`; one Flutter-Linux window rendering the four state colors. (Feeds the G5 FFI demo `StatusChip` `Connecting → Handshaking → Connected(masque,23ms)` [03-CLIENT §13 T0.1].)
- **T0.2 ConnectButton + StatusChip v0.** Driven *only* by the Rust event stream (CI2); the G5 demo's visible surface.

Phase 1 — `helix_design` v1 [03-CLIENT §13 T1.2]

- **T1.1 OpenDesign integration (§11.4.162).** Resolve D-DS-1 (the OpenDesign package/export, §14), wire `gen_tokens` codegen + drift gate (§1.3); migrate `HelixTokens` to the generated source.
- **T1.2 The five signature components**, each light+dark + every-state golden (DS-I2), the lint rules (§1.4), accessibility (§8).
- **T1.3 `colorForStatus`/`cueForStatus`** exhaustive over the doc-03 FFI `TunnelStatus`; the FFI contract test asserts the shared subset matches `02–ORCH §4.1` byte-for-byte (§4.1).
- **T1.4 AdaptiveScaffold** + breakpoints; compact/medium/expanded goldens.
- **T1.5 Motion + reduced-motion** (§7); ShieldIndicator honest-cost notes (§5.4).

Phase 2 — parity surfaces [03-CLIENT §13 T2.x]

- **T2.1 ExitPicker multi-hop** chain builder + jurisdiction labels (overlap-free, DS-I3).
- **T2.2 Direct-vs-relay `path` chip** — surface the `Connected.path` field once the core enum extension lands (§4.1 cross-doc seam).
- **T2.3 DAITA + PQ shield badges** wired to `Shields`.

Phase 3 — extended reach

- **T3.1 HarmonyOS/Aurora visual parity** — the UI ports for free; golden the same matrix on the fork builds [03-CLIENT §13 T3.x]; the design system needs **no** per-platform code (the shim does, not the widgets).
- **T3.2 Console web density** — deferred-loaded heavy widgets, ⌘K palette, expanded multi-pane goldens.

13. Cross-document contracts this document fixes

Contract	Fixed value	Consumed by
Connection-state palette (`stateDisconnected/Connecting/Connected/Danger` tokens + hex)	§1.2, §2.1	every screen; the white-label token-swap path
`colorForStatus`/`cueForStatus` over the doc-03 FFI `TunnelStatus`	§2.2	every state-bearing widget
Brand seed `0xFF3B5BDB`, light+dark first-class, per-role override seam	§1.2, §3.3	`helixTheme(flavor, brightness)`, all three flavors
`HelixTheme` ThemeExtension + `context.helix` accessor	§3.1	every widget; the DS-I1 no-hardcode rule
Signature component API (`ConnectButton/StatusChip/ExitPicker/ShieldIndicator/AdaptiveScaffold` signatures, callbacks-out rule)	§5	the three apps' screens [03-CLIENT §6/§8]
Breakpoint set (compact<600 / medium 600–1024 / expanded>1024) + branch-on-size rule	§6	`AdaptiveScaffold`, all responsive layouts
CI5 announce-not-color discipline + the §1.4 lints	§2.2, §8	every state widget; the a11y release gate
`TunnelStatus` consistency with `02-ORCH §4.1` (5-variant core subset byte-identical; 3 client extensions FFI-synthesized)	§4.1	the FFI contract test [03-CLIENT §12]; doc 02 (core enum)

14. Open decisions surfaced by this document

Per §11.4.6/§11.4.66 — options + recommendation, never silently resolved.

#	Decision	Option A	Option B	Recommendation
D-DS-1	OpenDesign (§11.4.162)	OpenDesign exports tokens (JSON/Style-		**A (codegen) —** keeps `HelixTokens` a `const` zero-

#	Decision	Option A	Option B	Recommendation
	Flutter binding	Dictionary-style) → `gen_tokens` → `tokens.g.dart` (codegen seam, §1.3)	OpenDesign ships a Flutter package consumed at runtime	runtime-cost surface (§9) and gives a build-time drift gate (§1.3). **Requires §11.4.99 latest-source verification of `github.com/nexu-io/open-design`'s actual export format before T1.1** — marked `UNVERIFIED` (§1.1) until then; extend OpenDesign upstream per §11.4.74 if its export lacks a token format.
D-DS-2	Per-role brand divergence	single seed `0xFF3B5BDB`, role distinguished by accent tint only	distinct per-role seeds (Access/Connector/Console)	**A** — the evidence base fixes one seed "overridden per role" without distinct values (§3.3); a tint shift preserves brand cohesion. Surface to the operator if a stronger per-role identity is wanted; the state palette is role-invariant either way.
D-DS-3	Brand typeface	a licensed brand sans (one weight family, subset)	a high-quality open font (Inter/IBM Plex) tokenized via OpenDesign	**B for MVP** — avoid Roboto-default look [04_UI §3.3] without licensing risk; swap to a licensed face later is a token change, not a refactor (DS-I1). `UNVERIFIED`: no specific typeface is

#	Decision	Option A	Option B	Recommendation
				named in the evidence base.
D-DS-4	Adaptive scaffold impl	hand-rolled `AdaptiveScaffold` (full control)	`flutter_adaptive_scaffold` package	**A** — full control over the connect-screen layout + the no-overlap guarantee (DS-I3); the package is a viable accelerator if its breakpoints map to §6 [04_UI §7].

Sources verified

- `04_VPN_CLD/HelixVPN-helix-ui-Flutter.md [04_UI]` — §3 (design system: tokens, brand seed `0xFF3B5BDB`, connection-state palette, signature components), §4 (Riverpod, the pure-function-of-stream rule), §7 (responsive breakpoints, AdaptiveScaffold), §9 (a11y: announce-not-color, light/dark, i18n), §10 (budgets), §12 (widget/golden testing).
- `final/03-client-core-and-ui.md [03-CLIENT]` — §0.1 (CI2/CI5 invariants), §3 (FFI `TunnelStatus`/`Shields`/`ExitOption` mirror), §4–§5 (`TunnelPlatform` + per-platform shims, the Danger/revoked source), §6 (flavors), §7 (design-system overview this doc deepens), §8 (Riverpod data layer), §10 (budgets), §13 (phases).
- `final/v02-data-plane/orchestrator-and-state.md [02-ORCH]` — §4.1 (canonical 5-variant `TunnelStatus`), §4.3 (rtt re-emit delta), §4.4 (`Down.reason` stable prefixes), §5.1 (`OrchState` no public projection for Idle/ShuttingDown), §8 (kill-switch/DNS sub-machine = the `Danger{kind}` source).
- `final/v02-data-plane/transport-trait.md [02-TX]` — §2 (`Transport::kind()` strings the `StatusChip` shows), §5.2 (the consumed `TunnelStatus` projection), §9 (the §11.4.169 test-type vocabulary this doc's §11 mirrors).
- `v09-research/_SYNTHESIS.md [SYN]` — §2 (Flutter + Rust-core floor), §4 (phases), §5 (`helix_design` system + signature components + connection-state palette), §9 (constitution bindings).
- `[05_YBO]` operator mandate — eight-platform requirement; tiny/fast/stable bar; responsive web.

Constitution: §11.4.44 (revision header), §11.4.6/§11.4.66 (UNVERIFIED marking; decisions = options + recommendation), §11.4.162 (OpenDesign mandatory design system, light+dark per component, no overlap, visual *regression), §11.4.169 (comprehensive test-type coverage), §11.4.5/§11.4.69/ §11.4.107/§11.4.158/§11.4.159/ §11.4.163 (captured / window-scoped / vision-validated evidence), §11.4.28/§11.4.74 (decoupled reusable package, extend-upstream), §11.4.151/§11.4.155 (release/recording prefixes), §1.1 (paired mutations on every lint/gate).*

End of nano-detail specification — `helix_design` design system (Volume 4, Clients). Deepens [04_UI §3] / [03-CLIENT §7]; consumes the FFI `TunnelStatus` contract [03-CLIENT §3] consistent with the core enum [02-ORCH §4.1]. Surfaced decisions: D-DS-1 (OpenDesign binding), D-DS-2 (per-role brand), D-DS-3 (typeface), D-DS-4 (adaptive scaffold) — all presented, none silently resolved (§11.4.66).